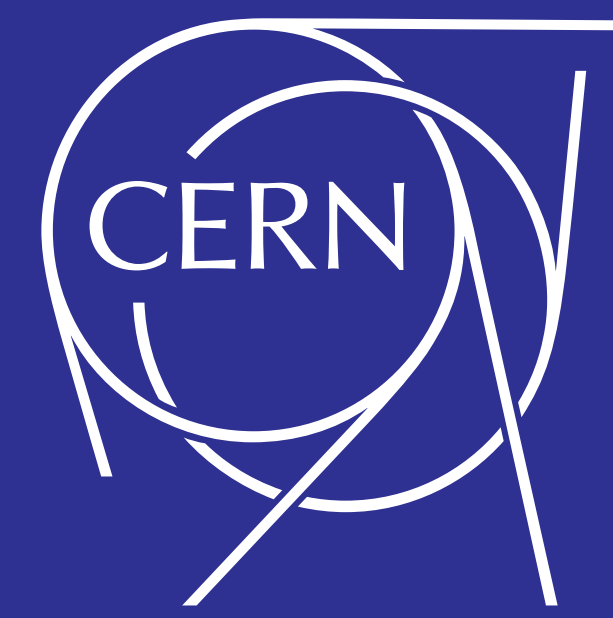




Poster ID:75

Geant4-driven realistic waveform generation for FPGA algorithm verification in radiation detector readout

Handan Yilmaz



Motivation

- FPGA-based radiation detector readout systems increasingly rely on real-time digital pulse processing for timing extraction, pile-up handling and neutron/gamma discrimination [1–5].
- Firmware verification is often performed using idealized, analytical or manually generated test pulses, which may not fully represent realistic detector-signal conditions.
- In practical readout chains, pulse-shape variability, electronic noise, baseline drift, ADC quantization and pile-up can shift timing estimates, distort pulse amplitudes and affect algorithm decisions.
- Physics-driven event simulation and detector-signal modeling provide repeatable waveform inputs with truth labels for HDL-level benchmarking, regression testing and corner-case analysis [1,3–5].

Key idea: convert Geant4 event information into realistic, label-aware ADC waveforms that can be reused as FPGA verification vectors.

Objective

This work demonstrates a Geant4-to-HDL verification workflow that transforms event-level deposited-energy information into realistic digitized waveforms for firmware testing.

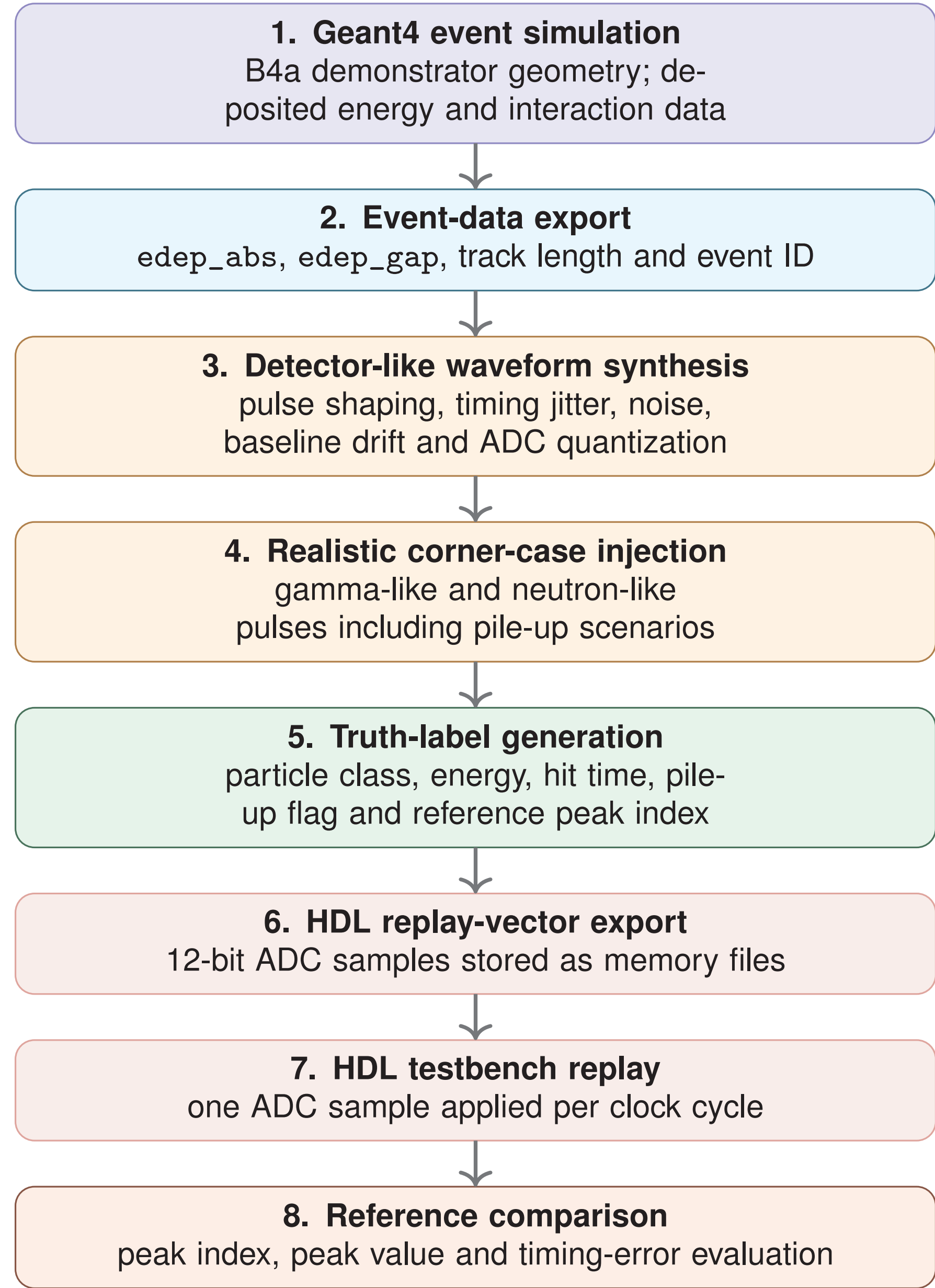
- Geant4 B4a is used as a controlled demonstrator event source.
- Deposited-energy data are converted into gamma-like and neutron-like ADC waveforms.
- Detector-response and electronics effects are introduced, including timing jitter, electronic noise, baseline drift, ADC quantization and pile-up.
- Each waveform is stored with truth labels, including hit time, deposited energy, particle class and pile-up flag.
- The generated traces are exported as 12-bit HDL replay vectors for testbench-based verification.

Contribution: a reusable simulation-to-HDL workflow for testing FPGA readout algorithms under realistic signal conditions.

Scope: this is a verification-oriented demonstrator, not a fully validated neutron/gamma detector-response model.

Geant4-to-HDL Workflow

A controlled Geant4 B4a demonstrator was used to generate event-level deposited-energy information. The proposed chain converts these events into realistic, label-aware ADC waveforms and reuses them as HDL verification vectors.



Workflow output: physics-driven ADC vectors, truth labels and reference metrics for repeatable HDL regression testing.

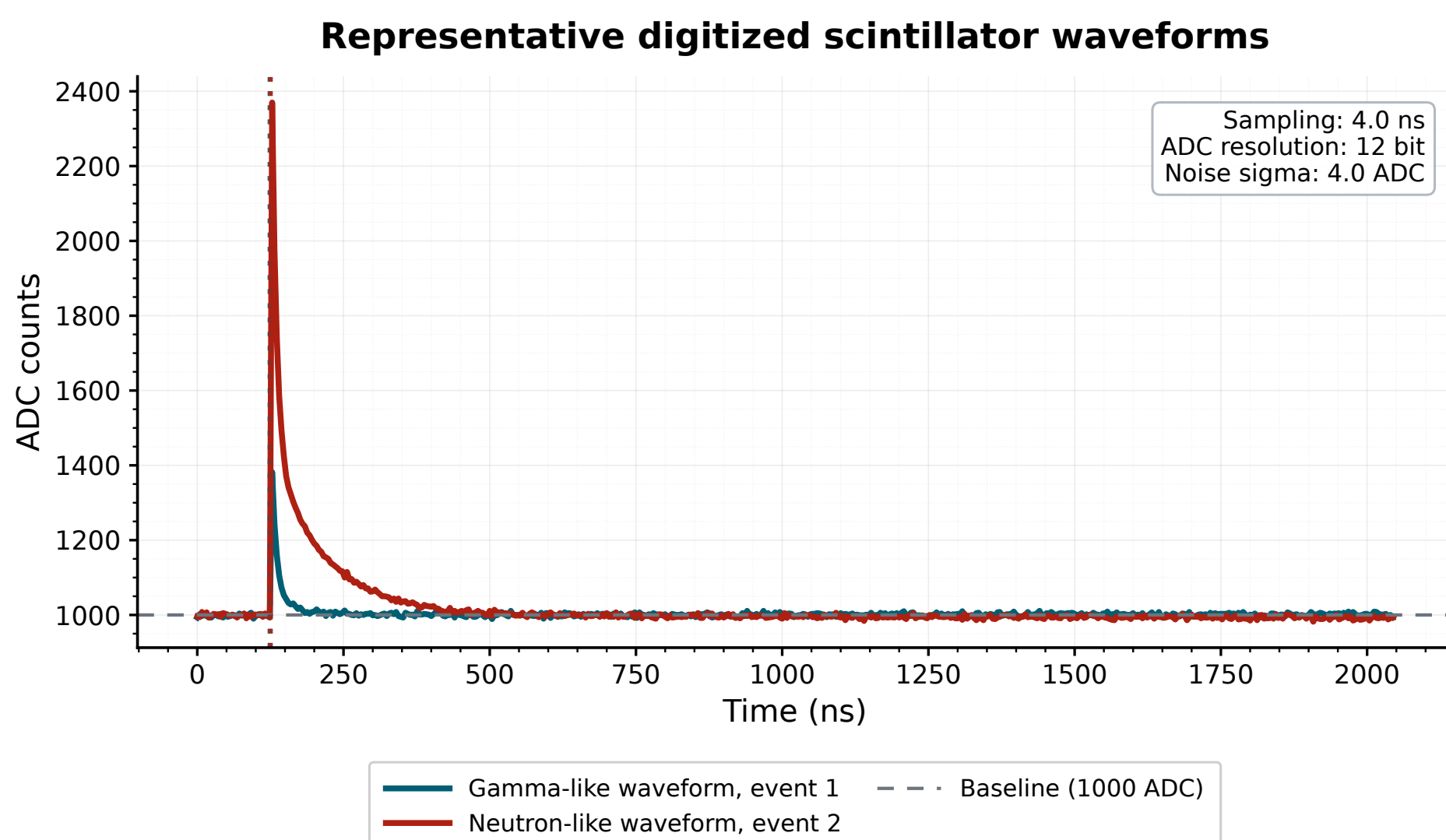
What this enables: the same waveform set can be reused to test timing extraction, peak detection and pile-up sensitivity under controlled but realistic signal conditions.

Verification focus: realistic waveform replay helps expose firmware mismatch cases before FPGA deployment.

Generated Dataset

A dataset of realistic ADC waveforms was generated from Geant4 event-level deposited-energy information. The dataset was designed to include both nominal pulses and challenging verification cases.

- Total events:** 2000 generated waveforms.
- Event classes:** 1000 gamma-like and 1000 neutron-like waveforms.
- Sampling:** 512 samples per event with 4 ns sampling interval.
- Digitization:** 12-bit ADC representation including baseline, noise and quantization.
- Pile-up cases:** 164 events include temporally overlapping pulses.
- Truth labels:** particle class, deposited energy, hit time, pile-up flag and reference peak index.



Representative generated gamma-like and neutron-like ADC waveforms.

HDL Replay and Verification

The generated waveform dataset was exported as HDL-compatible memory vectors and replayed in a testbench environment. Each ADC sample was applied sequentially to emulate a digitized detector-readout stream.

- Each event was stored as a 12-bit ADC memory vector.
- The HDL testbench replayed one ADC sample per clock cycle.
- A reference model was used to compute the expected peak index and peak value.
- HDL outputs were compared with the reference results on an event-by-event basis.
- Timing differences were evaluated using the sampling interval of 4 ns.

Verification step	Implementation
Input format	12-bit ADC waveform vectors
Replay mode	one sample per clock cycle
Reference quantities	peak index and peak value
Error metrics	index difference and timing error
Verification target	mismatch and corner-case detection

Verification goal: identify timing and amplitude mismatches that may be hidden when only idealized test pulses are used.

Results

The HDL replay results were compared against the reference model for all generated waveform events. The comparison confirms that the dataset can be used for repeatable firmware-level regression testing and corner-case analysis.

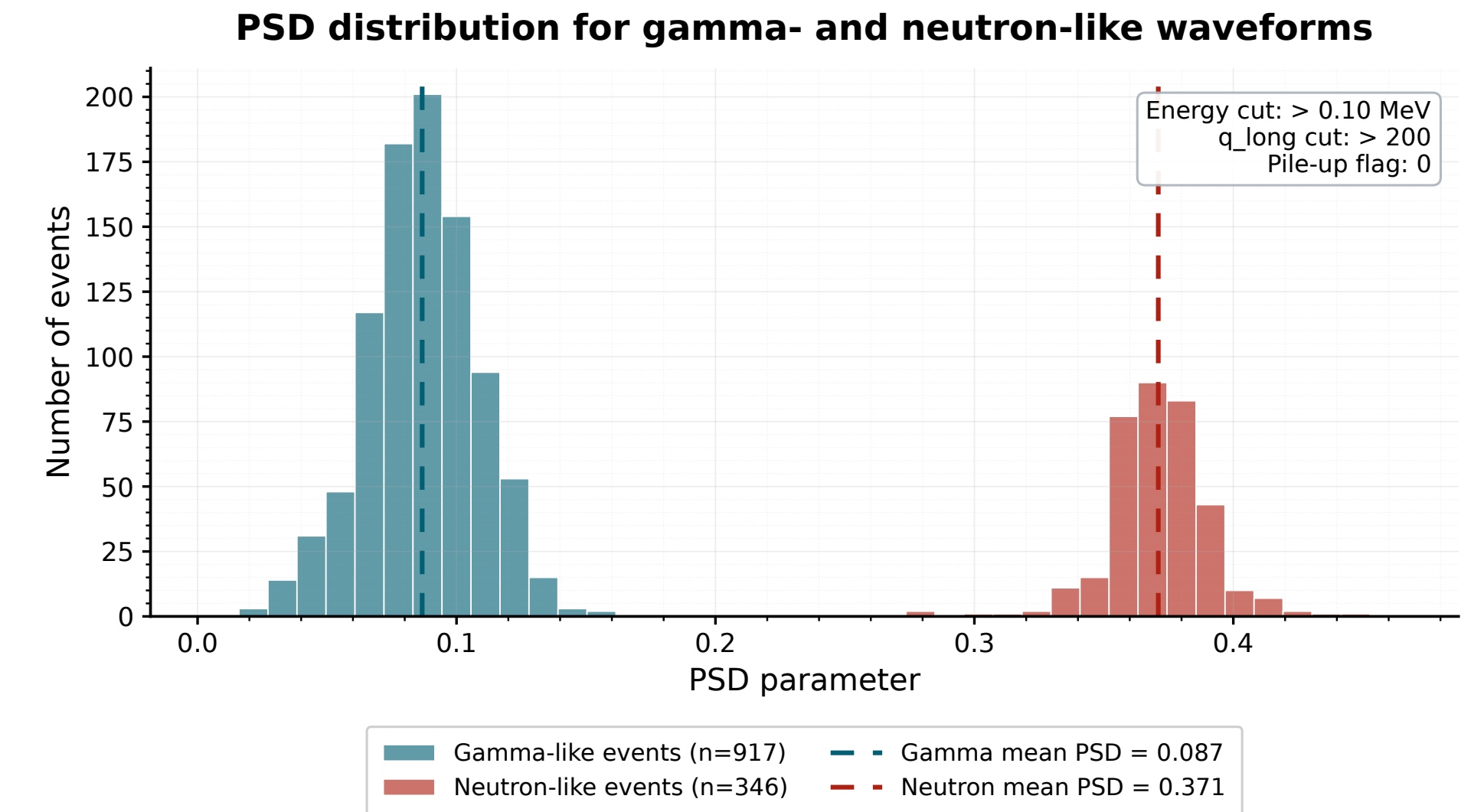
2000 compared events	96.95% exact peak-index match
1939 exact peak-index matches	4.45 mean index difference
89.97 ns mean timing error	164 pile-up events compared

Timing category	Mean error
Gamma-like events	2.46 ns
Neutron-like events	177.48 ns

- Most events show exact agreement between HDL replay and the reference model.
- Larger timing deviations are mainly associated with neutron-like pulse shapes and pile-up cases.
- These mismatch cases are useful for stress-testing FPGA readout algorithms before hardware deployment.

PSD Demonstrator

In addition to peak-based replay verification, the generated waveform set was also used to test a simple charge-comparison PSD demonstrator. This provides an additional check that the synthetic gamma-like and neutron-like pulses preserve distinguishable pulse-shape behavior.

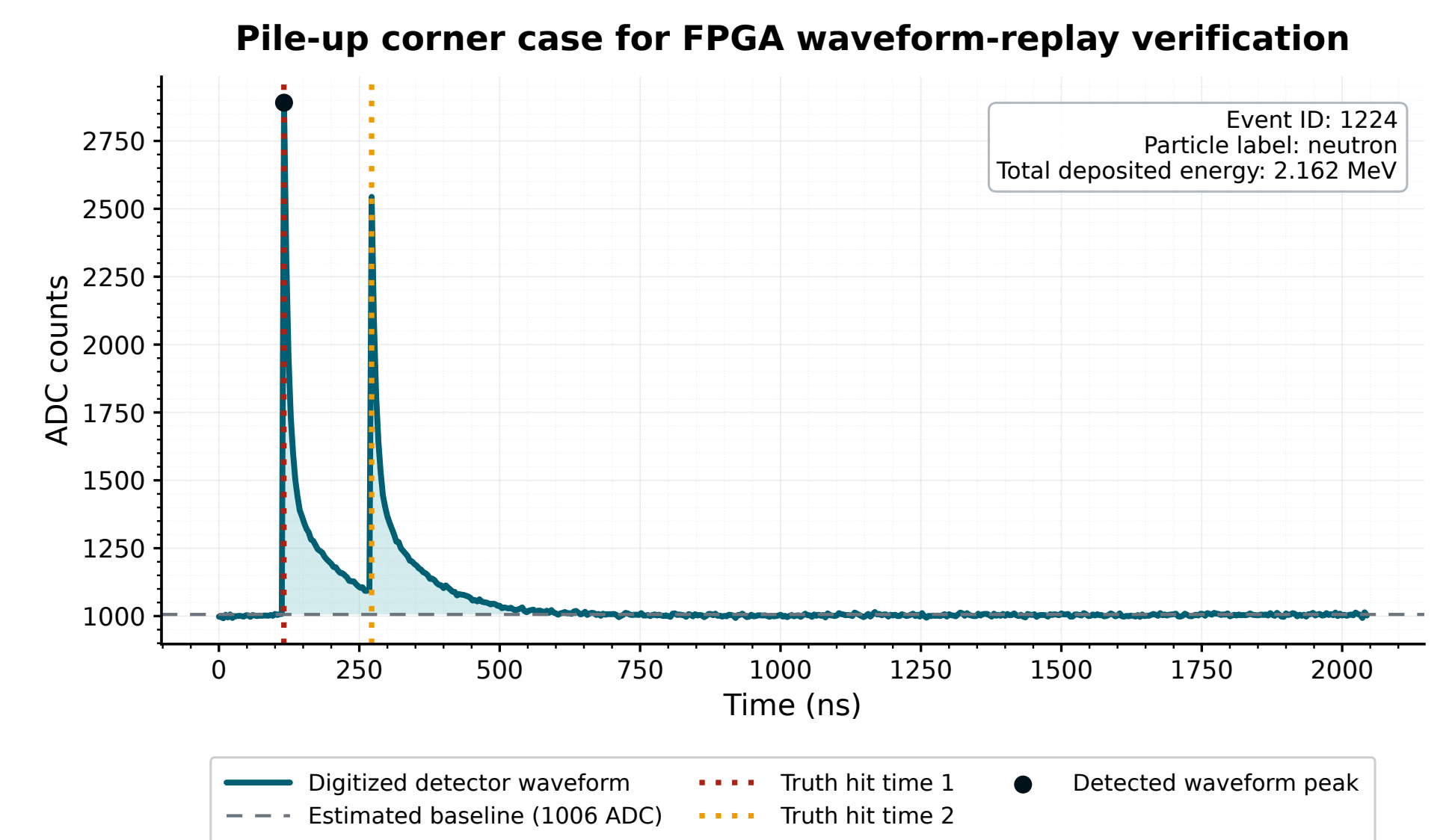


Distribution of the PSD parameter for generated gamma-like and neutron-like waveform classes.

- Gamma-like waveforms form a lower PSD-parameter population.
- Neutron-like waveforms show a larger tail-charge contribution and higher PSD values.
- The separation supports the use of the dataset for future FPGA-ready neutron/gamma discrimination studies.

Purpose: PSD is used here as a demonstrator metric, not as a fully validated detector-discrimination model.

Pile-up Corner Case



A representative pile-up waveform is shown as a timing-sensitive verification case. Dashed markers indicate the true hit times of two temporally close interactions.

- Overlapping pulses can shift the detected peak index.
- The combined waveform may distort the apparent pulse amplitude and timing.
- Such cases are difficult to reproduce with idealized single-pulse test inputs.

Why it matters: pile-up replay allows the HDL testbench to exercise realistic corner cases before FPGA deployment.

Summary and Outlook

Summary

- Geant4 event information was converted into realistic, label-aware ADC waveform vectors.
- The generated vectors were exported as HDL-compatible replay inputs.
- HDL outputs were compared with reference quantities such as peak index, peak value and timing error.
- The workflow provides physics-driven test inputs for FPGA firmware verification.

Main observation

- Idealized pulses can miss timing shifts caused by realistic noise, baseline drift and pile-up.
- Realistic waveform replay exposes mismatch cases earlier in the firmware-development cycle.

Next steps

- Improve detector-response and scintillation-pulse modeling.
- Evaluate FPGA-ready neutron/gamma discrimination algorithms.
- Study fixed-point precision, latency and resource trade-offs.

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MORE INFORMATION



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